Answers and talk notes

Math is Ingenuity

What your student is learning

Tonight we use ratios to explain how bicycle gears work — why one turn of the pedals can spin the wheel more than once.

Answers

Ask your student to explain before you look.

1. Adding a chain and gears so one pedal turn spins the wheel farther is ingenuity.
2. One full turn of the 48-tooth pedal gear is one rotation.
3. The 32-tooth gear on the back wheel is a gear. Its teeth catch the chain.
4. 48 pedal teeth to 32 wheel teeth is a ratio, written 48:32.
5. 48:32 and 3:2 are equivalent. Divide both by 16.

Talk about it

Who can have a clever idea about a problem?

Home mission

Find two jar lids, one big and one small. Stand them on the table with their rims touching. Put a sticky note on the top of each. Turn the big lid around once. Ask your student to predict first: will the small lid turn more than once?

When they are stuck, ask

- How many teeth does each gear have?
- What happens with one full turn of the pedals?
- Can you write that comparison as a ratio?

Try not to say

- Just add the teeth together.
- Bigger gear always means faster.
- Ratios are only for recipes.